

LearnSphere: Personalized Learning Management Platform

ABSTRACT

LearnSphere is a Spring Boot web application providing a comprehensive Learning Management System. It integrates student profiles, course video playlists, MCQ-based quizzes with auto-grading, academic result tracking with GPA/CGPA calculation, and role-based dashboards for Students, Faculty, and Administrators.

Built with Spring Boot 3.2.5, Thymeleaf, Spring Security 6, and H2 database, it automates academic management and provides performance insights through a responsive dark-themed interface.

INTRODUCTION

Traditional academic management relies on fragmented manual systems for course delivery, student assessments, and performance tracking. Students lack centralized platforms for accessing courses, tracking performance, and taking assessments.

LearnSphere addresses these challenges by providing an integrated web-based platform unifying learning resources, assessments, and academic analytics into one role-based system.

OBJECTIVES

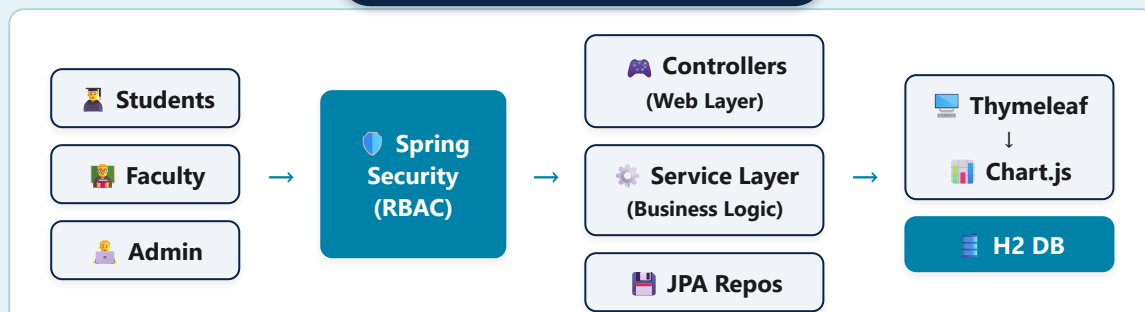
- Develop a unified Spring Boot platform for learning and academic management.
- Implement role-based access for Students, Faculty, and Admin with Spring Security.
- Enable MCQ quiz creation with auto-grading and performance analytics.
- Build academic results dashboard with GPA/CGPA calculation.
- Provide course video playlist management with YouTube integration.

METHODS & MATERIALS



- Framework:** Spring Boot 3.2.5, Java 17.
- Frontend:** Thymeleaf + Bootstrap 5.3 + Chart.js 4.
- Database:** H2 In-Memory (JPA / Hibernate).
- Security:** Spring Security 6, BCrypt.
- Build:** Apache Maven 3.9 / IntelliJ IDEA.

SYSTEM ARCHITECTURE



Layered MVC Architecture with Role-Based Security

RESULTS

Metric	Result
JPA Entities	12 Domain Models
Source Files	35 (0 Compilation Errors)
Startup Time	9.9 seconds
Roles	3 (Student / Faculty / Admin)
Courses & Videos	5 Courses, 27 YouTube Embeds
Quizzes & Questions	4 Quizzes, 29 MCQs
Academic Records	60 Results (3 Semesters, 15 Subjects)
Grade Scale	O, A+, A, B+, B, C, F with GPA

KEY FEATURES

- ✓ **Unified Course Management:** Centralized learning platform.
- ✓ **MCQ Auto-Grading:** Instant feedback and evaluation.
- ✓ **GPA Analytics:** Semester-wise performance dashboards.
- ✓ **Role-Based Access:** Secure portals for all stakeholders.
- ✓ **Video Integration:** Embedded YouTube course playlists.

CONCLUSION

LearnSphere successfully delivers a complete LMS platform integrating course management, quiz assessment with auto-grading, and academic analytics with GPA/CGPA computation.

Built on Spring Boot's layered MVC architecture with Spring Security for role-based access, it compiles 35 files with zero errors and starts in under 10 seconds, providing an engaging learning experience.

FUTURE ENHANCEMENTS

- Migrate to MySQL/PostgreSQL for persistent storage.
- Integrate AI-powered quiz generation from course content.
- Implement real-time video conferencing (WebRTC).
- Develop Android companion app using Kotlin.

REFERENCES

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